

AI Analytics Command Center

Agentic AI • Data Analytics • Automation • Modern Data Stack

Job-Ready Advanced Project Blueprint

A portfolio project designed for Data Analyst → Data Scientist → AI/Agentic AI career progression.

1. Project Vision

Build an enterprise-style analytics platform where a user can upload or connect business data and ask questions in natural language. The system autonomously validates data, performs SQL and Python analysis, detects anomalies, identifies likely root causes, generates business recommendations, and distributes automated reports through n8n.

Core outcome

Raw Data → Data Quality → SQL/Python Analysis → Anomaly Detection → AI Insights → Recommendations → Dashboard → Automated Report

2. High-Level Architecture

Layer	Technology	Purpose
Data Sources	CSV / Excel / JSON / APIs / DB	Raw business data
Storage	AWS S3	Raw/processed cloud data storage
Warehouse	PostgreSQL → Snowflake	Analytical storage and scalable SQL
Transformation	SQL + dbt	Clean, model and transform data
Analytics	Python, Pandas, NumPy, SciPy	EDA and statistical analysis
AI Agents	LLM API + LangGraph	Reasoning, routing and tool use
Automation	n8n	Scheduled pipelines, alerts and reporting
Backend	FastAPI	APIs and agent orchestration interface
BI	Power BI	Executive dashboards and KPI monitoring
Frontend	React / Next.js	Professional user interface

3. Agentic AI System

- 1. Data Quality Agent** — Profiles schema, checks missing values, duplicates, invalid dates, datatype issues, negative values and suspicious records. Produces a Data Quality Score and issue report.
- 2. Data Analyst Agent** — Understands the dataset and business context, selects useful KPIs, generates analytical questions and creates SQL tasks.
- 3. SQL Agent** — Generates SQL, executes it against the database, validates results and returns structured analytical outputs.
- 4. EDA / Statistics Agent** — Uses Python/Pandas/SciPy for distributions, correlation, trends, outliers and statistical tests where appropriate.
- 5. Anomaly Detection Agent** — Detects unusual changes in revenue, orders, returns, customer activity or other KPIs and ranks severity.
- 6. Root Cause Agent** — Breaks anomalies down by region, product, customer segment, channel or time to identify likely contributors.
- 7. Business Insight Agent** — Converts analytical results into concise, evidence-backed business insights.
- 8. Recommendation Agent** — Produces prioritized actions with rationale, expected impact and confidence level.
- 9. Report Agent** — Creates an executive summary for dashboards, email or scheduled reports.

4. Natural-Language 'Ask Your Data' Workflow

Example question: “**Why did revenue fall last month?**”

Question → Intent Agent → SQL generation → SQL execution → Period comparison → Breakdown analysis → Anomaly check → Root-cause analysis → Recommendation → Final answer

The UI should allow the user to expand the response and inspect the SQL used, source metrics, calculations and supporting charts. This makes the system more trustworthy than a generic chatbot.

5. n8n Automation Layer

Workflow	Trigger	Actions
Daily ingestion	Schedule	Fetch API/JSON → validate → store → trigger analysis
Analysis pipeline	New data	Quality Agent → SQL Agent → EDA → anomaly analysis
Critical alert	High anomaly	Generate explanation → send email/Slack/notification
Weekly executive report	Schedule	Calculate KPIs → AI summary → generate report → distribute

6. Product / Dashboard Pages

- Executive Overview — Revenue, profit, orders, customers, growth and major alerts.
- AI Insights — Automatically generated findings with evidence and confidence.
- Anomalies — Timeline of detected anomalies, severity and affected dimensions.
- Ask Your Data — Natural-language analytics with SQL transparency.
- Recommendations — Prioritized business actions and expected impact.
- Data Quality — Schema, missingness, duplicates, validation results and quality score.
- Agent Activity — Which agent ran, which tools were used, execution status and timestamps.
- Reports — Previous automated reports and export options.

7. Recommended Technology Stack

Category	Recommended
Programming	Python
Analytics	Pandas, NumPy, SciPy, scikit-learn, Plotly
Database	PostgreSQL
Cloud storage	AWS S3
Data warehouse	Snowflake (advanced version)
Transformation	SQL + dbt
Agent framework	LangGraph
LLM	One primary LLM API: OpenAI / Gemini / Claude
Automation	n8n
Backend	FastAPI
BI	Power BI
Frontend	React / Next.js
Version control	Git + GitHub

8. Development Roadmap

Phase 1 — Analytics Engine

Python + SQL + PostgreSQL + Power BI. Build cleaning, KPI calculations, EDA and dashboard.

Phase 2 — AI Layer

Implement Data Quality, SQL, EDA, Anomaly, Insight and Recommendation agents with tool access.

Phase 3 — Ask Your Data

Natural-language question → intent → SQL → execution → evidence → answer.

Phase 4 — n8n

Automate ingestion, scheduled analysis, alerts and weekly reports.

Phase 5 — Cloud

Move raw files to S3 and optionally warehouse analytical data in Snowflake.

Phase 6 — Production Polish

Authentication, logging, retries, agent traces, error handling, monitoring and documentation.

9. Dataset Recommendation

Use a realistic business dataset rather than a tiny classroom CSV. Good domains include e-commerce, food delivery, retail, SaaS subscriptions or restaurant operations. A strong version can combine orders, customers, products, payments, geography and returns. Start locally with a manageable dataset and design the pipeline so it can scale to millions of records.

10. Example AI Executive Output

■ BUSINESS PERFORMANCE

Revenue: ■48.2L

MoM Growth: -8.4%

■ KEY ALERTS

- North-region revenue ↓ 18%
- Product X returns ↑ 24%
- New customers ↓ 12%

■ KEY INSIGHT

The revenue decline is concentrated in the North region and is strongly associated with Category B performance.

■ RECOMMENDED ACTION

Investigate inventory availability and pricing for Category B in the North region.

Confidence: 87%

11. Resume-Ready Project Description

AI Analytics Command Center | Agentic AI • Python • SQL • Power BI • n8n • FastAPI • PostgreSQL • AWS S3

- Built an agentic analytics platform that automates the end-to-end data analysis workflow from ingestion and data-quality validation to SQL analysis, statistical exploration, anomaly detection and business recommendations.
- Developed specialized AI agents for data profiling, SQL generation, exploratory analysis, anomaly investigation and executive insight generation using tool-based workflows.
- Integrated n8n to orchestrate automated data ingestion, scheduled analysis, anomaly alerts and AI-generated business reports.

- Implemented a natural-language “Ask Your Data” interface that converts business questions into executable SQL, validates results and produces evidence-backed insights and recommendations.
- Integrated Python/Pandas, PostgreSQL and Power BI for data processing, analytical modeling and interactive business intelligence.

12. What Makes This Project Job-Ready

- Shows real Data Analyst fundamentals: SQL, Python, statistics, KPIs and visualization.
- Demonstrates modern data engineering concepts: cloud storage, warehouse, transformation and orchestration.
- Demonstrates practical Agentic AI: agents use tools and follow controlled workflows rather than only generating text.
- Demonstrates automation with n8n instead of manual report generation.
- Provides a strong interview story: business problem → architecture → tools → trade-offs → results.
- Can be presented as a scalable product rather than a one-off notebook.

13. Important Implementation Principles

- **Do not let the LLM invent numbers.** Metrics must come from SQL/Python/database results.
- **Validate generated SQL.** Use permissions, query limits and schema-aware tools.
- **Keep an audit trail.** Store agent, tool, query, timestamp and result metadata.
- **Separate reasoning from execution.** Agents decide what to do; trusted tools execute the actual computation.
- **Build V1 first.** Do not start with S3 + Snowflake + dbt + LangGraph + n8n + frontend all at once.

Recommended build order: Analytics Engine → Agentic Layer → Ask Your Data → n8n Automation → Cloud/Snowflake → Production polish.